

DMIT2015 – Lesson 4B Student Practice Activity

Introduction to Jakarta Faces (JSF)

Overview

In this activity, you will:

- Create a simple Jakarta Faces page
- Create a managed/backing bean
- Bind JSF UI components to Java properties
- Deploy and test a JSF application using Apache Tomcat

This activity reinforces:

- Facelets pages
- JSF components
- Expression Language (EL)
- managed beans
- server-side rendering

Learning Outcomes

By completing this activity, you should be able to:

- Create a Facelets page
- Create a CDI managed bean
- Bind UI components to Java code
- Deploy a Jakarta EE web application
- Test JSF interactions in a browser

Part 1 – Create the Backing Bean

Step 1 – Create Package

Create:

```
dmit2015.view
```

Step 2 – Create GreetingBean.java

Create the following Java class:

```
package dmit2015.view;

import jakarta.enterprise.context.RequestScoped;
import jakarta.inject.Named;
```

```

@Named
@RequestScoped
public class GreetingBean {

    private String firstName;

    public String getFirstName() {
        return firstName;
    }

    public void setFirstName(String firstName) {
        this.firstName = firstName;
    }

    public String getGreetingMessage() {

        if (firstName == null || firstName.isBlank()) {
            return "";
        }

        return "Welcome " + firstName + " to DMIT2015!";
    }
}

```

Part 2 – Create the JSF Page

Step 1 – Create index.xhtml

Create:

```
src/main/webapp/index.xhtml
```

Step 2 – Add the Following Code

```

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml"
      xmlns:h="jakarta.faces.html"
      xmlns:p="primefaces">

    <h:head>
        <title>DMIT2015 JSF Demo</title>
    </h:head>

    <h:body>

        <h2>DMIT2015 Jakarta Faces Demo</h2>

        <h:form>

            <p:messages />

            <p:inputText
                value="#{greetingBean.firstName}"

```

```
        placeholder="Enter your first name" />

        <p:commandButton
            value="Submit"
            update="@form" />

        <br/>
        <br/>

        <h:outputText
            value="#{greetingBean.greetingMessage}" />

    </h:form>

</h:body>

</html>
```

Part 3 — Run the Application

Instructions

1. Start Apache Tomcat
2. Deploy the application
3. Open a browser
4. Navigate to:

<http://localhost:8080/project-context-path>

Part 4 — Verify Functionality

Test the Following

Test 1

Enter:

Sam

Expected output:

Welcome Sam to DMIT2015!

Test 2

Refresh browser page.

Verify:

- form reloads
 - no server errors occur
-

Part 5 — Modify the Application

Task 1 — Update the Page Title

Change:

DMIT2015 Jakarta Faces Demo

to your own custom title.

Task 2 — Add Another Output Message

Add:

```
<h:outputText value="Apache Tomcat + Jakarta Faces" />
```

below the greeting message.

Bonus Challenge (Optional)

Add:

- lastName property
- second input field
- full greeting message

Example:

Welcome Sam Wu to DMIT2015!

Common Troubleshooting

Problem

Cannot resolve symbol jakarta.faces.html

Solution

Refresh Maven dependencies.

Problem

Blank page or HTTP 404

Solution

Verify:

- Tomcat started successfully
- application deployed successfully

- correct URL used
-

Problem

EL expression not working

Solution

Verify:

- bean uses:

@Named

- getter/setter names match property names
-

Submission

No Brightspace submission required unless instructed by your instructor.

Be prepared to:

- demonstrate your JSF page
- explain how binding works
- show successful deployment in browser